

Qatar-2b

R-0757

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_240421 · created 2026-07-28T03:37:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460421.8407 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.021
Transit depth (Rp/Rs)^2	2.89 +/- 0.71 [%]
Orbital Inclination (inc)	89.74 +/- 1.53
Airmass coefficient 1 (a1)	1.023 +/- 0.025
Airmass coefficient 2 (a2)	-0.017 +/- 0.03
Scatter in the residuals of the lightcurve fit is	2.45 %
Best Comparison Star	#2 - [172, 184]
Optimal Method	PSF photometry
Transit Duration (day)	0.0768 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.17 ± 0.021 vs 0.16327 (deviation 0.32σ)
Duration fit vs published	0.0768 d vs 0.0767 d (deviation 0.07σ)
Mid-time O-C	-0.7 min
Depth SNR	8.1
Residual scatter	2.45 %

Light curve

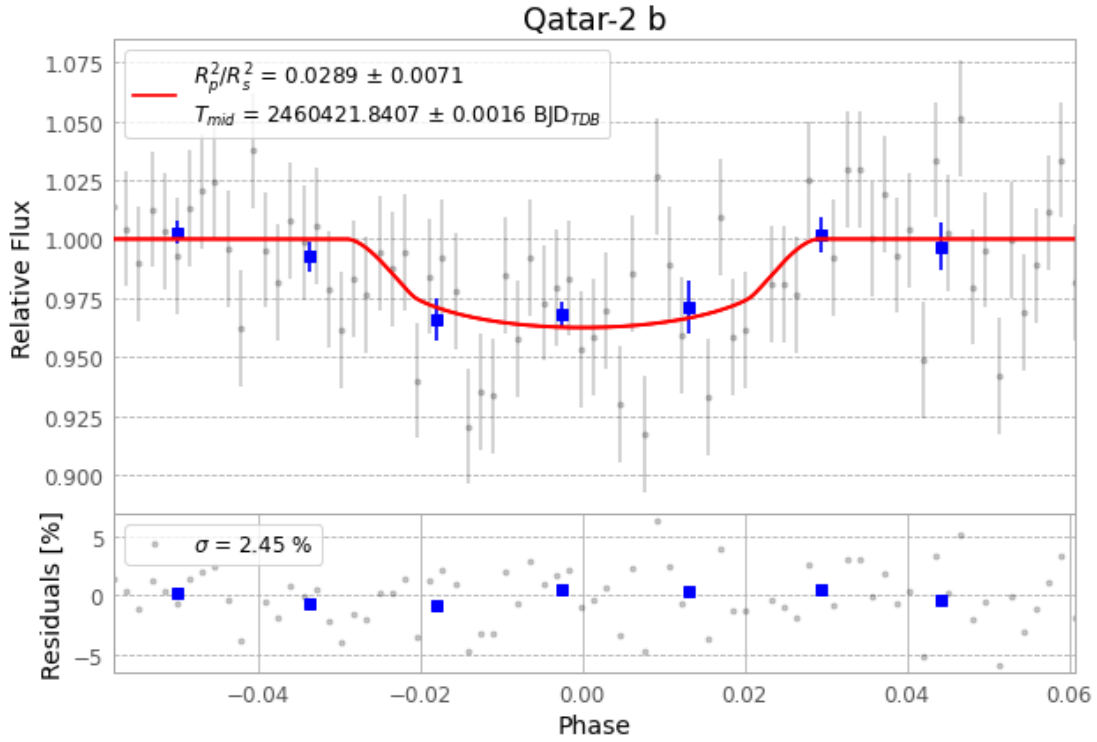


Figure 1 — FinalLightCurve_Qatar-2 b_2024-04-20.png (SHA-256 066b2d38008c6c55...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [523, 122], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2b5f87756780b301...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), Qatar-2240421060918.FITS → Qatar-2240421095715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-240421.log (e638fab8f2d4...), FinalLightCurve_Qatar-2 b_2024-04-20.png (066b2d38008c...), FinalLightCurve_Qatar-2 b_2024-04-20.csv (8cfb7422cdbe...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 03:37Z.